
Data Visualization

Telling Your Story with Data

AUC Data Science Initiative | Data Challenge Workshop

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Agenda

- Why Visualization Matters & Know Your Audience
- Building a Narrative Arc
- Good vs. Bad Visualizations
- Choosing the Right Chart + Visual Gallery
- Chart Deep Dives with Python Code
- Real Data Walkthrough (Case Study)
- Python Libraries & Color Theory
- Design Best Practices & Common Pitfalls
- Data Challenge Tips & Q&A

Why Visualize?

- Humans process visuals 60,000x faster than text
- A good chart can replace paragraphs of explanation
- Visualization reveals patterns, trends, and outliers hidden in raw data
- Your audience remembers what they SEE, not what they read

Know Your Audience

Academic / Professor

- Expects methodological rigor
- Show statistical context
- Label everything precisely
- Include sample sizes and sources

General Audience

- Keep it simple and visual
- Use familiar chart types
- Lead with the insight, not the method
- Minimize jargon

Data / Technical Team

- Can handle complexity
- Show distributions, not just averages
- Include confidence intervals
- Interactive plots welcome

Ask yourself: "Who is looking at this, and what do they need to understand?"

Turning Analysis into a Story

Every great analysis tells a story. Your visualizations should map to a narrative arc.

Setup

What is the context? What question are we answering?

Conflict

What does the data reveal?
What is surprising or important?

Resolution

What should we do about it?
What are our recommendations?

Every Chart Should Have a Job

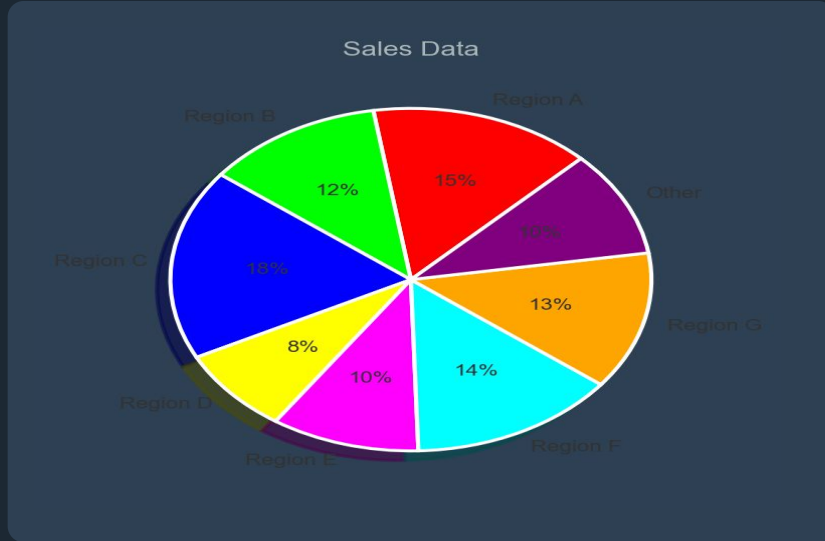
- Before making a chart, ask: "What is this supposed to show?"
- Each visualization should directly support a key finding
- If a chart doesn't advance your story, cut it
- Label charts with insight titles, not just variable names

Bad: "Sales by Region"

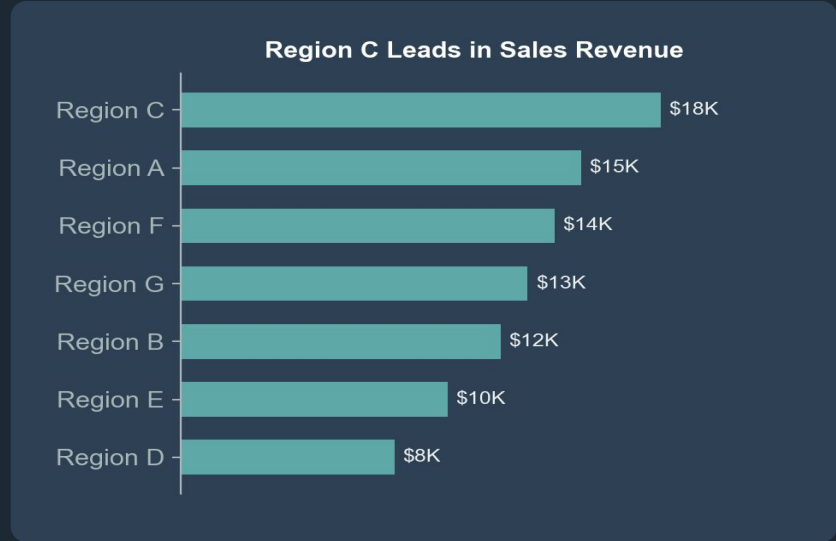
Good: "The Southeast Region Leads All Markets in Q4 Growth"

Good vs. Bad: See the Difference

Don't Do This



Do This Instead



- Remove 3D effects, excessive colors, and chartjunk
- Use clear titles that state the insight

Before & After: Slide Design

Before

DATA ANALYSIS RESULTS FOR STUDENT SURVEY Q4 2025 FINDINGS AND CONCLUSIONS

We looked at the data from the survey and found that students in the Southeast region had the highest scores. The average was 3.8 and the standard deviation was 0.4. We also looked at other regions and found similar patterns but with lower scores. The Northeast had 3.2 and the Midwest had 3.5. We ran a t-test and found p less than 0.05.

In conclusion there are many differences between regions and more research is needed to fully understand the implications of these findings for future policy decisions and academic programs.

After

Southeast Students Outperform All Regions

Average GPA: 3.8 vs 3.2-3.5 in other regions



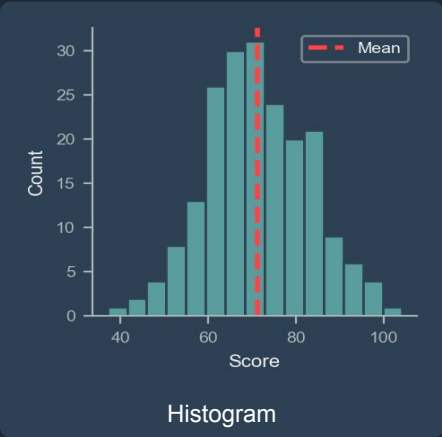
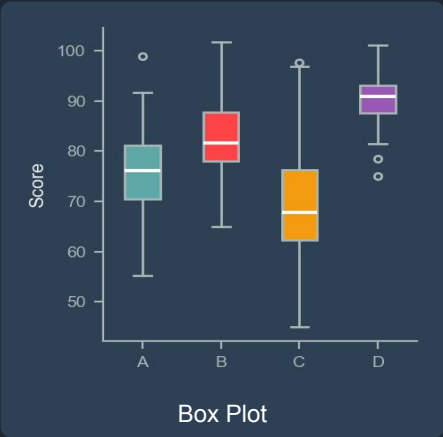
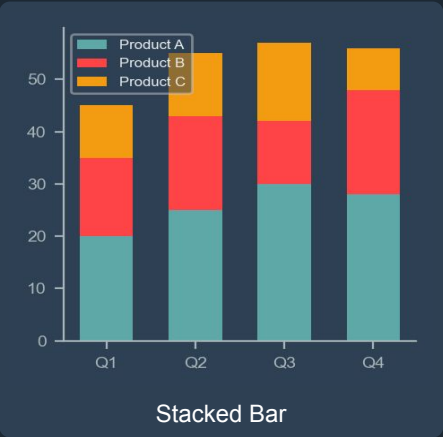
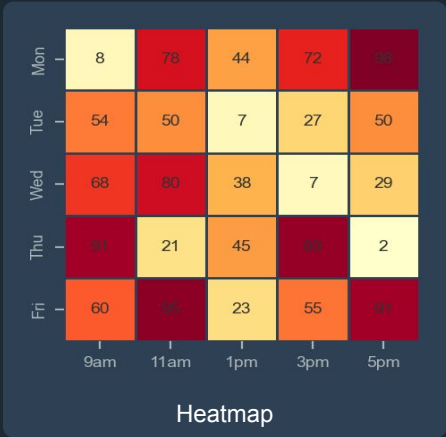
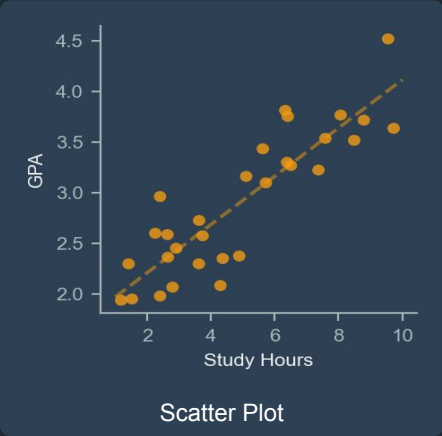
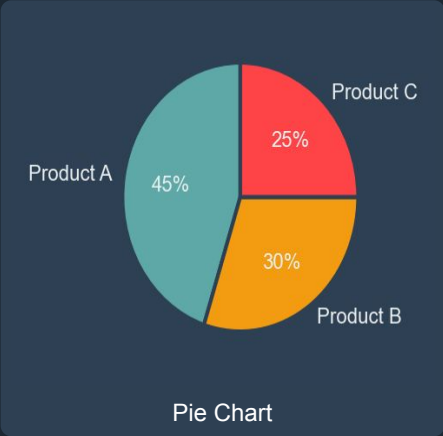
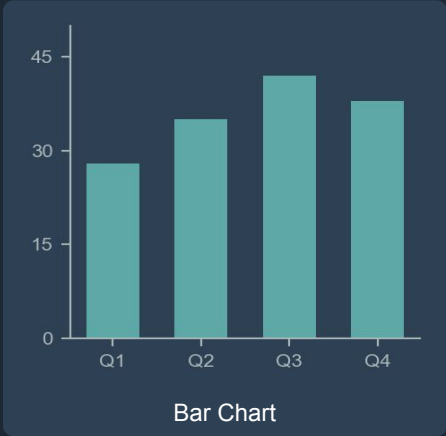
Lead with the insight. Let the chart do the heavy lifting.

Choosing the Right Chart

Match the chart type to the message

Your Goal	Chart Type
Comparison	Bar Chart
Trend Over Time	Line Chart
Proportion	Pie Chart
Composition Across Groups	Stacked Bar
Distribution Shape	Histogram
Compare Distributions	Box Plot
Relationship	Scatter Plot
Patterns Across 2 Dimensions	Heatmap

Chart Type Gallery



Comparison: Bar Charts

Best for: comparing categories side by side

WHEN TO USE

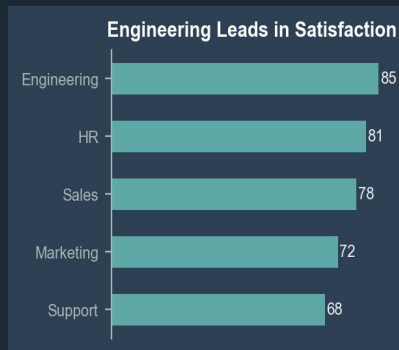
Comparing values across groups: sales by region, scores by department, enrollment by major. The audience instantly sees who's ahead.

- Sort bars by value (not alphabetical) to reveal ranking
- Horizontal bars for long category labels
- Use one color; highlight the key bar if needed
- Title should state the insight, not just variables

STORY ROLE

"Here's who's winning and by how much" -- perfect for your Conflict or Resolution section.

DO



Sorted, insight title, clean

DON'T



Rainbow, unsorted, vague title

Trends: Line Charts

Best for: showing change over time

WHEN TO USE

Tracking metrics over time: revenue by month, enrollment by year, temperature by day. The audience sees direction and momentum.

- Limit to 2-3 lines max -- more gets confusing
- Annotate key moments (peaks, drops, crossovers)
- Use markers sparingly for data points
- Always include a clear legend if multiple lines

STORY ROLE

"Here's how things changed over time" -- ideal for showing trends in your Setup or Conflict.

DO



2 lines, annotated, insight title

DON'T



7 lines, no legend, vague title

Composition: Pie & Stacked Bar

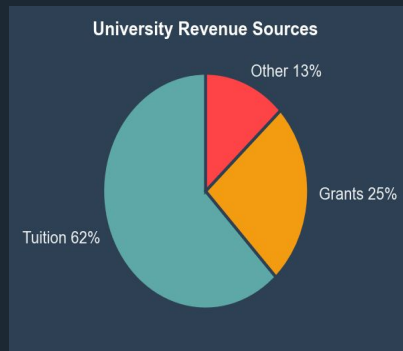
Best for: showing parts of a whole

WHEN TO USE

Showing proportions: budget breakdown, market share, demographic splits. The audience sees how pieces add up to 100%.

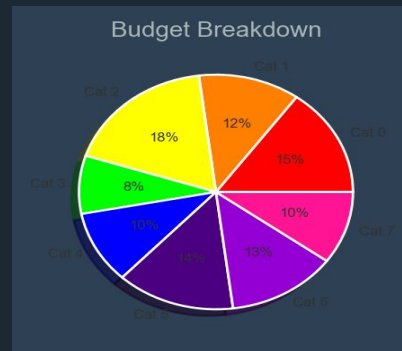
- Pie: ONLY 2-4 slices. More? Use a bar chart
- Always label with percentages, not just names
- Stacked bars better for comparing across groups
- Never use 3D or "exploded" pie charts

DO



3 slices, labeled, clear title

DON'T



8 slices, rainbow, unreadable

STORY ROLE

"Here's where the money goes" -- great for framing context in your Setup section.

Distribution: Histograms & Box Plots

Best for: understanding how data is spread out

WHEN TO USE

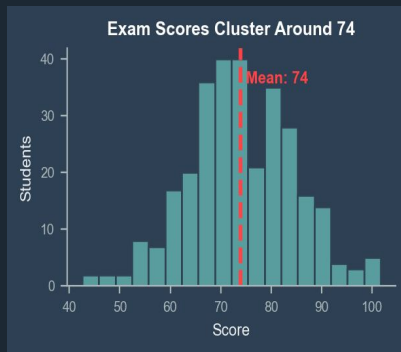
Exploring spread and shape: exam score distributions, salary ranges, response times. Shows clustering, skew, and outliers.

- Histograms: choose bins carefully (too few = hiding shape)
- Add a mean/median line for quick reference
- Box plots: great for comparing groups side by side
- Label axes -- "Count" and "Score" not just numbers

STORY ROLE

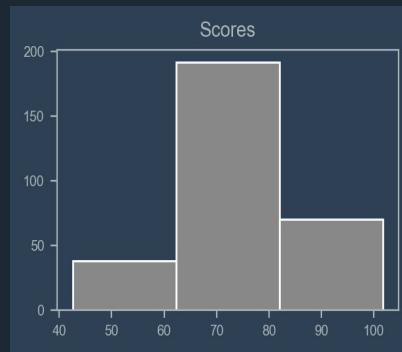
"Here's the full picture, not just the average" -- reveals depth in your Conflict section.

DO



Good bins, mean line, labeled

DON'T



3 bins, no labels, no insight

Relationships: Scatter Plots

Best for: exploring correlations between two variables

WHEN TO USE

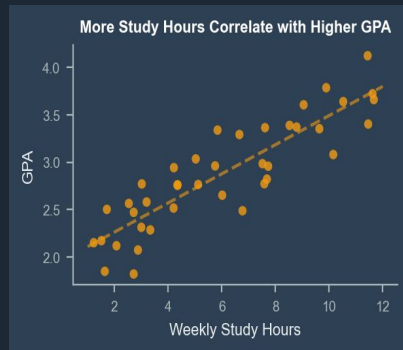
Testing relationships: study hours vs GPA, spending vs revenue, temperature vs energy use. Does X actually affect Y?

- Add a trend line to show direction of relationship
- Use color or size to encode a third variable
- Label both axes with units
- Don't overplot -- if too dense, use transparency

STORY ROLE

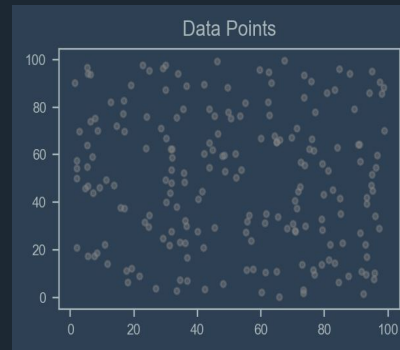
"Here's what's actually driving results" -- the evidence behind your recommendations.

DO



Trend line, labeled, insight title

DON'T



Random data, no labels, no story

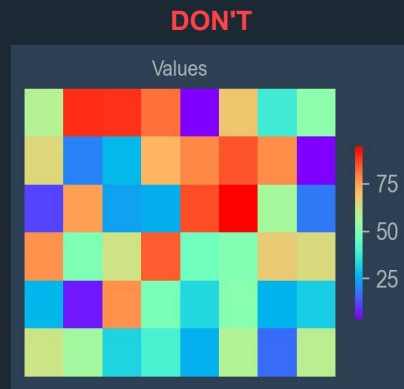
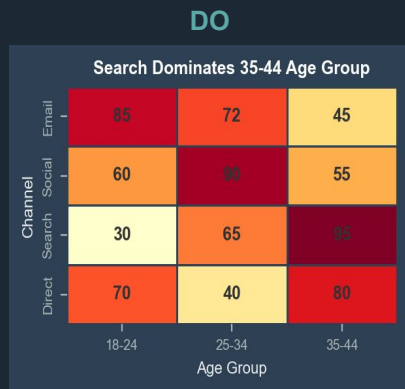
Patterns: Heatmaps

Best for: spotting patterns across two categorical dimensions

WHEN TO USE

Finding hotspots in two-dimensional data: website traffic by day and hour, engagement by channel and age group, correlation matrices.

- Use sequential colormaps (YlOrRd, Blues) not rainbow
- Annotate cells with values for precision
- Keep the grid small -- too many cells gets unreadable
- Label both axes clearly with meaningful categories



STORY ROLE

"Here's where the action is" -- reveals hidden patterns to guide decisions in your Resolution.

CASE STUDY

Real Data Walkthrough: The Question

"Which major has the highest average GPA?"

Our Dataset (sample)

Major	Students	Avg GPA
Biology	142	3.45
Comp Sci	98	3.21
Psychology	167	3.62
Business	134	3.18
Education	89	3.71

Step 1: Start with a clear, specific question your data can answer.

CASE STUDY

Choosing the Right Chart

We are comparing categories (majors)

Bar Chart is the right choice

Step 2: Think about your goal

Comparing categories = Bar Chart

Trend over time = Line Chart

Part of whole = Pie Chart

The Code

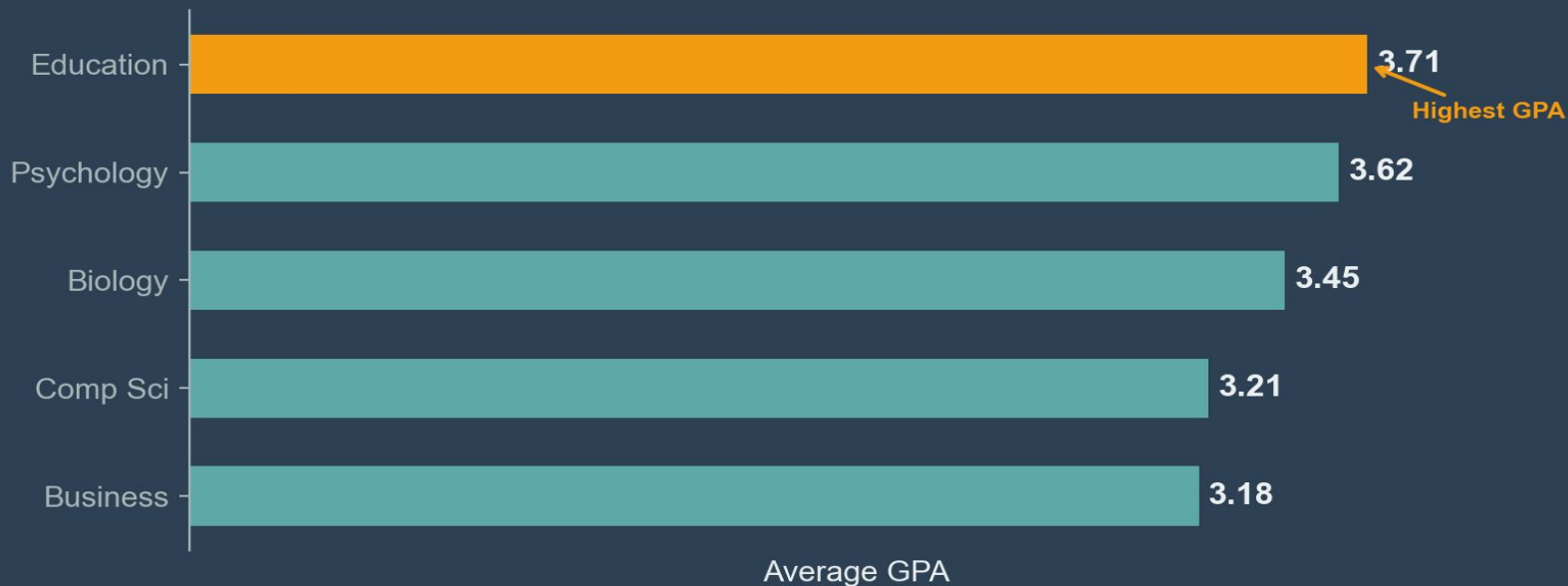
```
import seaborn as sns
import matplotlib.pyplot as plt

# Sort by GPA for clarity
df_sorted = df.sort_values(
    'avg_gpa', ascending=True
)

sns.barplot(
    data=df_sorted,
    x='avg_gpa',
    y='major',
    palette='viridis'
)
```

CASE STUDY

Education Majors Lead in GPA Across All Programs



Step 3: Use an insight title. Let the chart tell the story.

Notice: sorted bars, clean labels, descriptive title -- not just "GPA by Major"

Python Libraries for Data Visualization

matplotlib

The foundation. Full control, more verbose.

Best for customized static plots.

seaborn

Built on matplotlib. Beautiful defaults, great for statistical plots.

Less code needed.

plotly

Interactive charts. Great for dashboards and exploration.

Hover, zoom, filter built in.

From Data to Viz in 5 Lines

Complete Example

```
import seaborn as sns
import matplotlib.pyplot as plt

# Load a built-in dataset
tips = sns.load_dataset('tips')

# Create visualization
sns.barplot(
    data=tips, x='day', y='total_bill',
    hue='sex', palette='Set2'
)
plt.title('Average Bill by Day and Gender')
plt.tight_layout()
plt.savefig('my_chart.png', dpi=150)
```

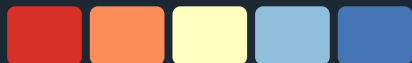
Color Theory for Data Viz

Sequential



Low to high values (e.g., temperature, intensity)

Diverging



Two extremes with a neutral midpoint (e.g., positive/negative)

Categorical



Distinct groups with no inherent order (e.g., regions, majors)

Colorblind-Safe



Accessible to ~8% of men with color vision deficiency

- In seaborn: `palette='viridis'` (sequential), `'Set2'` (categorical), `'coolwarm'` (diverging)
- Tools: colorbrewer2.org | seaborn.pydata.org/tutorial/color_palettes

Accessible & Inclusive Design

Making your visualizations work for everyone

~8% of men and ~0.5% of women have some form of color vision deficiency. If 50 people see your chart, chances are 2-4 can't read it the way you intended.

BEST PRACTICES

- Never use color alone to encode meaning -- pair with shapes, patterns, or labels
- Avoid red/green together -- the most common type of colorblindness
- Use high contrast between text and background (4.5:1 ratio min)
- Add alt text to charts in presentations and reports
- Test your palette: simulate what colorblind viewers see
- Use large enough fonts -- 12pt min for slide body text

Colorblind-Friendly Palettes

viridis



cividis



Color Universal



Blue-Orange



Tools to Check Accessibility

- Coblis -- Color Blindness Simulator (color-blindness.com)
- Viz Palette -- susie-lu.github.io/viz-palette
- In Python: `palette='viridis'` or `palette='cividis'`
- WebAIM Contrast Checker (webaim.org/resources/contrastchecker)

Designing Clean, Intentional Visuals

- Less is more -- remove chartjunk (unnecessary gridlines, borders, 3D effects)
- Use color with purpose -- highlight what matters, gray out what doesn't
- Choose accessible color palettes (colorblind-friendly)
- Use consistent fonts and sizing
- White space is your friend
- Every element should earn its place on the chart

Labels Matter

- Always include: title, axis labels, units
- Use descriptive titles that state the insight, not just the variables
- Annotate key data points to guide the viewer's eye
- Add data source and context when presenting to others
- Legend placement: don't let it compete with the data

Visualization Pitfalls to Avoid

- Pie charts with too many slices
- Truncated y-axis that exaggerates differences
- Using 3D charts (they distort perception)
- Rainbow color schemes (hard to read, not accessible)
- Too much data on one chart -- if it is confusing, split it
- Missing titles, labels, or context
- Choosing a chart type because it "looks cool" vs. because it fits

Storytelling Pitfalls to Avoid

- Dumping all your charts without a narrative thread
- Showing every analysis you ran -- only show what supports your story
- Forgetting your audience -- tailor the level of detail
- Presenting findings without recommendations
- Burying the lead -- put your most important insight first

Key Takeaways

- Start with the story, THEN choose your charts
- Every chart should have a clear job and support a finding
- Match chart type to the message (comparison, trend, distribution, etc.)
- Design for clarity -- remove clutter, use color intentionally
- Use seaborn or plotly for fast, beautiful Python visualizations
- Practice! The more you visualize data, the better you get

Data Challenge Tips

How to make your presentation stand out

- 1 Lead with your most important insight**
Don't make judges wait -- put your best finding front and center
- 2 Every slide should advance the story**
Cut anything that doesn't directly support your narrative
- 3 Keep visualizations clean and labeled**
Judges should understand your chart in under 5 seconds
- 4 End with clear, actionable recommendations**
What should someone DO with your findings?
- 5 Practice your delivery out loud**
Time yourself. Know your transitions. Rehearse with your team.

Questions?

Resources to Keep Learning

- Seaborn Gallery: seaborn.pydata.org/examples
- Matplotlib Tutorials: matplotlib.org/stable/tutorials
- Plotly: plotly.com/python
- Book: "Storytelling with Data" by Cole Nussbaumer Knaflic
- Chart Chooser: data-to-viz.com